MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, POWER

## TYPE 8438 1/

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Tetrode

See figure 1

Mounting position: Vertical, base down or up

Weight: 9 ounces nominal

ABSOLUTE RATINGS: F1 = 110 MHz

| Parameter: | Ef | Eb | Ec1 | Ec2 | Ib | Pp | Pg1 | Pg2 | Cooling |
|---|---|---|---|---|---|---|---|---|---|
| Unit: | Vac | Vdc | Vdc | Vdc | mAdc | W | W | W | See note 1 |
| Maximum: | | | | | | | | | |
| C Telep: | 5.0 ±5% | 3,200 | -500 | 600 | 275 | 270 | 10 | 35 | --- |
| C Teleg: | 5.0 ±5% | 4,000 | -500 | 600 | 350 | 400 | 10 | 35 | --- |
| AB Audio: | 5.0 ±5% | 4,000 | -500 | 800 | 350 | 400 | 10 | 35 | --- |
| TEST CONDITIONS: | 5.0 | 2,500 | Adj | 500 | 160 | --- | --- | --- | See note 2 |

GENERAL:

Qualification - Required

1/ Formerly JAN 8438/4-400A



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following the applicable test.

| | METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| | | Quality conformance inspection, part 1 | | | | | | | |
| | 1301 | Filament current | | 0.65 | II | If | 13.5 | 14.7 | Aac |
| | 1266 | Total grid current | See note 4 | 0.65 | II | Ic1 | --- | -10 | μAdc |
| | 1266 | Primary-grid emission (control) | Ef = 5.5 Vac; Ic1 = 165 mAdc; t = 15; anode and screen grid floating | 0.65 | II | Isg1 | --- | -100 | μAdc |
| | 1266 | Primary-grid emission (screen) | Ef = 5.5 Vac; Ic2 = 135 mAdc; Ec1 = 0; t = 15; anode floating | 0.65 | II | Isg2 | --- | -200 | μAdc |
| | 1261 | Electrode voltage (grid) | | 0.65 | II | Ec1 | -55 | -80 | Vdc |
| | 1231 | Peak emission | eb = ec1 = ec2 2,500 v | 0.65 | II | Is | 7.0 | --- | a |
| | | Quality conformance inspection, part 2 | | | | | | | |
| | 1316 | Amplification factor | g1 to g2; Ic2 70 mAdc Eb 0 | --- | --- | Mu | 4.5 | 6.0 | --- |
| (B) | 1236 | Power oscillation | Class C amplifier; grid driven; Eb = 3 kVdc; Eg1/Ib = 350 mAdc; F = 110 MHz (min) | --- | --- | Po | 500 | --- | W (useful) |
| | 1331 | Direct-interelectrode capacitance | | --- | --- | Cgp<br>Cin<br>Cout | ---<br>10.70<br>4.20 | 0.17<br>14.50<br>5.60 | pF<br>pF<br>pF |
| | | Quality conformance inspection, part 3 | | | | | | | |
| | --- | Life test | Group C; power output, t = 500 hours | --- | --- | --- | --- | --- | --- |
| | --- | Life-test end points: | | | | | | | |
| | 1231 | Peak emission | | --- | --- | Is | 5.6 | --- | a |
| | 1261 | Primary-grid emission (control) | | --- | --- | Isg1 | --- | -100 | μAdc |
| | 1266 | Primary-grid emission (screen) | | --- | --- | Isg2 | --- | -200 | μAdc |
| (B) | | Periodic-check tests | | | | | | | |
| | 1031 | Low-frequency vibration | No voltages (see note 5) | --- | --- | --- | --- | --- | --- |
| | --- | Low-frequency vibration end points: | | | | | | | |
| | 1261 | Electrode voltage (grid) | | --- | --- | Ec1 | -55 | -80 | Vdc |
| | 1266 | Total grid current | | --- | --- | Ic1 | --- | -10 | μAdc |
| | 1042 | Shock, specified pulse | No voltages; accel = 15 G peak (min); D = 11 ±2 ms half-sine (see note 5) | --- | --- | --- | --- | --- | --- |



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|---|
| | | | | | | MIN | MAX | |
| | Periodic-check tests -Continued | | | | | | | |
| --- | Shock, specified pulse end points: | | | | | | | |
| 1261 | Electrode voltage (grid) | | --- | --- | Ec1 | -55 | -80 | Vdc |
| 1266 | Total grid current | | --- | --- | Ic1 | --- | -10 | µAdc |

NOTES:

1. Forced-air cooling shall be provided for the base-pin and anode-lead seals. This air should be applied simultaneously with filament power, using the 259-JAN socket. A minimum airflow of 14 cu. ft./minute shall be used. The pressure drop as measured in the socket at this flow equals 0.25 inches of water. The air requirements stated above are based on operation at sea level and on ambient temperature of 20°C. Operation at high altitude or at high temperature requires a greater volume of airflow.

2. Forced-air cooling of the anode and base seals to an extent not to exceed the minimum values specified in note 1 is permitted.

3. It is recommended that a heat-radiating type of connector (EIMAC TYPE HR-6, or equivalent) be used on the anode terminal in all rf applications.

4. This test is to be the first test performed at the conclusion of the holding period.

5. Shock and vibration tests shall be performed quarterly, with sampling as follows:

   n1 = 4 c1 = 0 Where c2 represents the total allowable failures for the first and
   n2 = 4 c2 = 1; second samples combined.

   Separate samples may be used at the option of the manufacturer, and the tests shall be considered as nondestructive except in case of failure. In case of failure after double sampling, the failing test shall become a quality conformance inspection, part 2, AQL 6.5 percent, inspection level S3, for three consecutive successful submissions, at which time the testing may revert to the periodic-check test basis.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Air Force - 11, 99
DSA - ES

User activities:
Army - AV, WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Army - EL

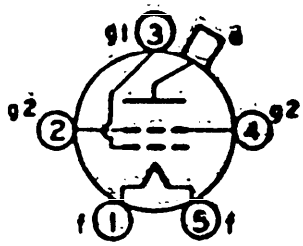
Agent: DSA - ES

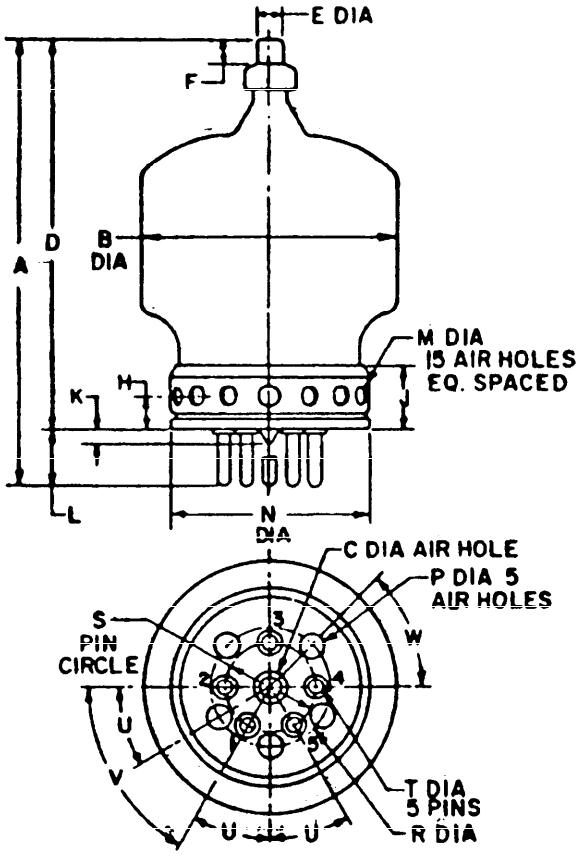
(Project 5960-2460)



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| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | | NOTES |
|---|---|---|---|
| | Minimum | Maximum | |
| Quality conformance inspection, Part 2 | | | |
| A | 5.875 (149.23) | 6.375 (161.93) | |
| B | | 3.563 (90.50) | |
| K | | .250 (6.35) | a |
| T | .185 (4.70) | .191 (4.85) | a |
| Quality conformance inspection, Part 3 (periodic check) | | | |
| D | 5.125 (130.18) | 5.625 (142.88) | |
| E | .350 (8.89) | .365 (9.27) | |
| F | .328 (8.33) | | |
| J | | .969 (24.61) | |
| N | | 2.750 (69.85) | |
| Reference dimensions | | | |
| C | .500 (12.70) | | |
| H | .438 (11.13) | | |
| L | .750 (19.05) | | |
| M | .250 (6.35) | | |
| P | .312 (7.92) | | |
| R | 1.625 (41.28) | | |
| S | 1.250 (31.75) | | a |
| U | 30° | | a |
| V | 60° | | a |
| W | 45° | | a |

NOTE:
a. Base pins T and tubulation K are so alined that they can be freely inserted in a gage 1/4 inch(6.35 mm) thick with hole diameters of .204(5.18 mm) and .500(12.70 mm), respectively; located on the true centers by the given dimensions S; U; V.

FIGURE 1. Outline drawing of electron tube type 8438.



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FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
ATTN: AMSEL- PP-ED
Fort Monmouth, New Jersey 07703

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| SPECIFICATION ANALYSIS SHEET | Form Approved<br>Budget Bureau No. 22-R255 |
|---|---|

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-E-1/887B ELECTRON TUBE, POWER TYPE 8438 1/

ORGANIZATION

| CITY AND STATE | CONTRACT NUMBER |
|---|---|
| | |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE |
|---|---|
| | |

DD FORM JAN 64 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-PM 1068-68

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